

BIHAR MSME POLICY AI

Immediate Development Sequence

Team Synopsis | Phase 1

PROJECT OBJECTIVE

Build a locally hosted AI policy intelligence system on the DGX Spark that can understand Bihar MSME Policy 2026, answer policy questions with citations, analyse investor proposals, calculate applicable incentives, identify missing information, and draft official responses.

Core principle

RAG	provides authoritative policy knowledge and citations.
Rule Engine	determines eligibility, exclusions and compatibility.
Calculation Engine	computes incentive amounts and caps.
LLM	understands the query, reasons over verified outputs and drafts the response.
DGX Spark	keeps the model and core processing local to the department environment.

Initial technology stack

DGX Spark | Qwen 30B/32B-class instruct model | FastAPI | React | PostgreSQL | Qdrant | Python

1 Core Project Setup

Purpose: Establish the working local application skeleton and model connectivity.

- Create separate modules for frontend, backend, LLM serving, RAG, policy engine, incentive engine, document parser, database, test cases and policy data.
- Keep services modular so policy knowledge, calculations and drafting can evolve independently.

Deliverable: Frontend -> backend -> local LLM communication working on the DGX Spark.

2 Convert the Policy into Structured Data

Purpose: Create a machine-readable policy specification instead of relying only on PDF chunks.

- Extract Policy -> Chapter -> Section -> Clause -> Rule -> Condition -> Source Page.
- Structure definitions, MSME classification, district categories, sectors, negative list, heritage clusters, incentives, caps, exclusions, claim conditions and policy-wide limits.
- Give special attention to eligibility, financial incentives and guiding conditions.

Deliverable: Versioned MSME Policy JSON plus PostgreSQL tables.

3 Build Citation-Backed Policy RAG

Purpose: Enable precise natural-language policy Q&A with source traceability.

- Ingest clause-level chunks into Qdrant with metadata: policy, version, section, page and topic.
- Use hybrid retrieval, query rewriting and reranking before sending context to the LLM.
- Never allow the model to answer policy facts without retrieved support when a source should exist.

Deliverable: MSME Policy chatbot returning answer + exact section/page citation.

4 Build MSME Eligibility Engine

Purpose: Determine formal eligibility through deterministic logic before any benefit calculation.

- Classify Micro / Small / Medium using applicable investment and turnover criteria.
- Determine new / existing / expansion, district Region A / B, sector class, negative-list status and promoter category.
- Return missing information rather than guessing when classification cannot be completed.

Deliverable: Structured eligibility API, e.g. POST /evaluate-eligibility.

5 Build Incentive Calculation Engine

Purpose: Convert each financial incentive into executable, testable rules.

- Implement capital subsidy, additional subsidy, scaling-up, payroll, power tariff, rooftop solar, audit support, stamp duty, quality, IP, SME exchange, SGST, e-commerce, export, revival and special packages.
- Apply caps, aggregate limits, exclusions and compatibility rules programmatically.
- The LLM explains the calculation; it does not perform or invent the governing arithmetic.

Deliverable: Structured benefit API, e.g. POST /calculate-benefits.

6 Policy Ambiguity & Safety Layer

Purpose: Prevent the system from inventing answers where the draft is incomplete, inconsistent or dependent on future guidelines.

- Store unresolved issues as structured flags requiring departmental clarification.
- When a rate, condition or interpretation is not supported, return "Policy clarification required" and do not assume a value.
- Version every clarification so future notified policy and amendments can supersede draft rules cleanly.

Deliverable: Policy ambiguity register plus response guardrails.

7 Investor Letter Analyzer

Purpose: Convert incoming letters/PDFs into a structured investment proposal.

- Extract company, investment, preferred district, sector, plant & machinery investment, turnover, land requirement, employment, promoter category and project type.
- Separate known information, missing information and assumptions.
- Support typed letters first; PDF/image ingestion can be added through the same extraction schema.

Deliverable: Investor extraction API, e.g. POST /analyse-investor-letter.

8 Unified Investor Analysis

Purpose: Connect extraction, eligibility, RAG, calculations and validation into one workflow.

- Show investment summary, enterprise classification, applicable policy provisions, eligible incentives, indicative benefit, ineligible benefits, missing information and policy clarifications.
- Every material policy conclusion should retain its source section/page.
- Keep calculation results machine-readable so they can be reused in reports and letters.

Deliverable: Complete investor-analysis screen and structured report.

9 Official Letter / Note Generation

Purpose: Draft departmental communication only from verified structured analysis.

- Provide the original investor letter + verified policy analysis + drafting instructions to the LLM.
- Do not allow the drafting layer to introduce new incentive amounts or unsupported commitments.
- Keep generated content editable and auditable before departmental use.

Deliverable: Editable Reply Letter, Internal Note, Director Brief and Investor Benefit Sheet.

10 Create Evaluation & Regression Dataset

Purpose: Test whether the system is consistently correct across realistic cases before expanding scope.

- Vary investment size, MSME category, Region A/B, sector type, promoter category, project type and incentive combinations.
- For every case store expected classification, eligibility, calculation, clause and response.
- Use failures and officer corrections as the foundation for later fine-tuning data.

Deliverable: At least 50 manually validated investor test cases with expected outputs.

Phase 1 Completion Target

Phase 1 is complete only when the following workflow works reliably end-to-end:

1. Investor Letter	2. Extract Project Details	3. Ask for Missing Information	4. MSME Classification	5. Policy Eligibility
6. Calculate Incentives	7. Check Caps / Exclusions	8. Retrieve Citations	9. Generate Analysis Report	10. Draft Official Reply

Immediate Scope Exclusions

Do not expand into the following areas until the MSME Policy 2026 intelligence layer is reliable:

- BIADA live land availability
- District-vs-district recommendation engine
- Sector-vs-district scoring
- GIS
- Other Bihar industrial policies and schemes
- Fine-tuning

Expansion Roadmap After Phase 1

Phase 2	BIADA land + district industrial data
Phase 3	Sector/location recommendation engine
Phase 4	Additional Industry policies and schemes
Phase 5	Cross-policy comparison and scheme stacking
Phase 6	Fine-tuning using officer-approved cases and corrections

TEAM RULE

**RAG provides policy knowledge. Rules determine eligibility. Code calculates benefits.
The LLM explains and drafts.**

Source basis: Draft of MSME Policy, Government of Bihar - Directorate of MSME, Department of Industries, Bihar (2026).